Ahsan Bilal

Reasoning & Test-Time Compute · Alignment & Post-Training · Efficient LLM Inference (KV Cache) · Agentic AI

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RESEARCH INTERESTS

Reasoning, Test-Time Compute Scaling, and RL *Papers:* ICML'26, ICLR'27, NeurIPS'26, EMNLP'26

Reasoning LLMs and inference-time (test-time) compute scaling: adaptive compute allocation, chain-of-thought, long-horizon and multi-step reasoning, process and outcome reward models (PRMs/ORMs), verifier-guided and tree/trajectory-level search (MCTS, best-of-N, self-consistency), self-correction and self-refinement, test-time training and adaptation, reasoning distillation, and reinforcement learning with verifiable rewards (RLVR; PPO, GRPO), for autoregressive and discrete (masked) diffusion language models, with uncertainty-aware exploration and test-time discovery.

Alignment, Preference Optimization, and AI Safety *Papers:* ICLR'27, COLM'26, EMNLP'26, TMLR'26

AI alignment and LLM post-training: RLHF, RLAIF, direct preference optimization (DPO and variants), general (non-Bradley–Terry) preference optimization, reward modelling, reward decomposition, reward hacking and overoptimization, constitutional AI, scalable oversight, and weak-to-strong generalization. AI safety, security, and reliability: sycophancy, hallucination and factuality, epistemic uncertainty and calibration, robustness, red teaming, jailbreaks and prompt injection, adversarial robustness, interpretability, trustworthy AI, and AGI safety (superalignment).

Agentic AI and Efficient LLM Systems *Papers:* ICLR'27, EAAI, ICC'25, KDD'26, ICASSP'26

LLM agents and multi-agent systems: tool use, planning, long-term memory, agent-to-agent communication protocols, reputation and trust, self-improving and self-evolving agents (self-reflection, self-play, recursive self-improvement), agentic RL, and agent safety. Memory and inference efficiency: KV cache eviction, compression, and quantization, sparse and head-adaptive attention, long-context modelling, memory optimization, speculative decoding, and efficient serving. Applications in multimodal RAG and ML-driven wireless communication (5G), grounded in optimization theory, statistics, and probabilistic modelling.

EDUCATION

University of Oklahoma, Oklahoma, USA 2024–Present

Ph.D. in Computer Science; M.S. en route (Deep Learning, RL, Reasoning & Alignment in LLMs)

- CGPA: 3.86/4.0
- Advisor: Dr. Dean Hougen: Research on deep learning optimization, robust LLM reasoning and alignment, memory-efficient inference, and RL-based agentic systems.

National University of Sciences & Technology (NUST-SEECS) Aug. 2020 – June 2024

Bachelor of Engineering in Electrical Engineering *Islamabad, Pakistan*

- CGPA: 3.66/4.0 | Specialization GPA: 4.0/4.0 | Merit Scholarship 2020
- Senior Design Thesis: Person identification using gait with fused graph and 3D-convolutional architectures (Presentation)

RESEARCH EXPERIENCE

REAL Lab, University of Oklahoma Graduate Researcher

Advisor: Dr. Dean Hougen *Aug 2024 – Present*

- Research on adaptive test-time compute for LLMs, including verifier/PRM-guided reasoning, agentic RL controllers, trajectory-level search, and dynamic compute allocation across autoregressive (ICML'26) and diffusion language models (ICLR'27). Written technical specifications and rigorous statistical analysis for each system; communicated findings cross-functionally across collaborating institutions following collaborative teamwork best practices.
- Work on structured generative models (discrete diffusion), test-time scaling during denoising, and continuous learning diffusion models for wireless channel prediction under distribution shift (ICASSP'26). Maintained well-documented, modular codebases; produced written technical documentation for all research deliverables.

- Research on memory- and compute-efficient LLM inference: Monte Carlo future-query estimation for KV cache eviction and long-context memory optimization (ICLR'27), and uncertainty-aware, self-correcting reasoning via refinement and disagreement distillation (EMNLP'26, NeurIPS'26).

Stanford AI Lab (SAIL), Stanford University Research Collaborator

Supervisors: Prof. John M. Cioffi, Dr. Emily Fox | Collaborator: M. A. Mohsin *Aug 2024 – Present*

- Developed diffusion-based wireless estimation and neural Gaussian radio fields (nGRF), including verifier-guided trajectory search for diffusion test-time scaling and adaptive compute allocation. Applied signal processing techniques and machine learning (ML) pipelines using pandas and scikit-learn for data preprocessing, classification tasks, and large-scale analysis.
- Worked on continuous learning under distribution shift and multimodal RAG pipelines for wireless systems. Collaborated with stakeholders from Stanford Statistics, Glasgow, and NUST in a cross-functional, collaborative research environment; publications in NeurIPS'25, ICML'25, AAAI'25, ICC'25 (Best Paper), KDD'26.
- Research on LLM alignment, preference optimization, and reliability: general preference reinforcement learning, continuous-utility DPO, sycophancy disentanglement via reward decomposition, and epistemic uncertainty for test-time discovery (ICLR'27, EMNLP'26, COLM'26).

Conoid.ai (Internet of Agents Platform) Research / Engineering

Ongoing

- Building a decentralized Internet of Agents platform for self-evolving, self-improving AI agents with persistent memory and agentic workflows, enabling dynamic agent teams based on trust, reputation, and prior performance. Designing scalable distributed system architecture and HTTP-based agent communication protocols with security built into all transactions; written system specifications covering all key use cases and acting as a self-starter driving product development across the full engineering flow.

Undergraduate Researcher Optimal ML Lab, NUST

Supervisor: Dr. Ahmad Salman *Jan 2023 – June 2024*

- Built robust CV/biometric models: boosted-attention ViT for shadow removal and fused GCN + 3D-CNN for gait recognition. Applied analytical methods with attention to detail to achieve state-of-the-art performance on benchmark datasets; supervised experimental pipelines from problem formulation through evaluation.
- Papers under review: Shadow Removal with Boosted Attention, Gait ID using Fused Graph + 3D-CNN.

INDUSTRY EXPERIENCE

Machine Learning Engineer Islamabad, Pakistan

Cowlar Design Studio (Y Combinator 17) — Based in USA *Feb 2024 – Aug 2024*

- Built an Action Recognition system for Smart Carts (95% accuracy) with dual-inference deployment across edge devices and Nvidia GPU clusters for scalable, real-time inference; developed clean, modular C++/Python components for high-throughput data pipelines.
- Automated fiber-cable alignment via computer vision and ML (96% success), improving precision to 5 μm and scaling production 40x; reported technical status and findings to cross-functional teams and stakeholders.
- Delivered AI-powered retail automation with Al Meera Consumer Goods Company, Qatar (∼QAR 2.9B+ revenue, ∼QAR 2.7B market cap), partnering with product and infrastructure teams; added security controls for high-volume transactions and Dash dashboards for monitoring.

SELECTED PUBLICATIONS

Full list on Google Scholar * indicates equal contribution; linked titles go to arXiv.

Efficient LLM Inference, Reasoning, and Test-Time Compute

[PC1] **What If We Allocate Test-Time Compute Adaptively?** ICML'26
A. Bilal*, A. Mohsin*, M. Umer, A. Subhan, H. Rizwan, A. Mohsin, D.F. Hougen

[PC2] **Monte Carlo Estimation for KV Cache Eviction** ICLR'27 (Submitted)
A. Bilal, M.A. Mohsin, M. Umer, W.H. Raza, A.U. Asad, Y.D. Kwon, M. Valko, D.F. Hougen

[PC3] **Attend at the Right Resolution: Head-Adaptive Sparse Attention via Query–Key Interaction Structure** ICLR'27 (Submitted)
W.H. Raza, A. Bilal, Y. Shen, R. Hu, X. Fu, Y. Wen

[PC4] **General Preference Reinforcement Learning** ICLR'27 (Submitted)
M. Umer, M.A. Mohsin, A. Bilal, E. Fox

[PC5] **AgentPersonaBench: Benchmarking Persona-Driven User Simulation** arXiv'26
J. Huang, Y. Wang, H. Shen, Y.D. Lu, S. Huang, M. Oh, Y. Wang, M.A. Mohsin, Z. Xu, Y. Fan, Z. Yuan, A. Bilal, et al. (46 authors) | Open-source contribution to MatrAIx.ai: dataset, environments, and evaluation code

[PC6] **Epistemic Uncertainty for Test-Time Discovery** ICLR'27 (Submitted)
K. Riaz, M.A. Mohsin, A. Bilal, M. Umer, A. Mohsin, A. Riaz, A. Subhan, J.M. Cioffi

[PC7] **S^3: Stratified Scaling Search for Test-Time in Diffusion Language Models** ICLR'27 (Submitted)
A. Bilal*, M.A. Mohsin*, M. Umer, D.F. Hougen

[PC8] **Faithfulness Is Not Free: Auditing Offline KV-Cache Quantization in Retrieval-Augmented Generation** EMNLP'26 Workshop
A.U. Asad*, A. Bilal*, M. Ali, M. Haseeb, D.F. Hougen

[PC9] **Distilling Disagreement at Test Time Reasoning** NeurIPS'26 (Submitted)
A. Bilal*, M.A. Mohsin*, M. Umer, E. Fox, D.F. Hougen

[PC10] **On the Fundamental Limits of LLMs at Scale** TMLR'26
A. Mohsin, A. Bilal, W. Zhao, M. Umer, et al. (with researchers from Google DeepMind and Meta)

[PC11] **Pressure, What Pressure? Sycophancy Disentanglement in Language Models via Reward Decomposition** COLM'26 (Submitted)
M.A. Mohsin*, A. Bilal*, M. Umer, E. Fox

[PC12] **Continuous-Utility Direct Preference Optimization** EMNLP'26 (Submitted)
M.A. Mohsin*, M. Umer*, A. Bilal, Z. He, M.U. Rafique, A. Aali, M.A. Jamshed, J.M. Cioffi, E. Fox

[PC13] **Refining Over Resampling: Test-Time Self-Correction for LLM Reasoning** EMNLP'26 (Submitted)
A. Bilal, M.A. Mohsin, M. Umer, L. Trigg, A. Subhan, M. Ali, D.F. Hougen

[PC14] **Adaptive Replay Refinement: Allocating Test-Time Compute Between Revision and Resampling** arXiv'26
A. Bilal, M.A. Mohsin, M. Umer, A. Mohsin, S. Bilal, D.F. Hougen

[PC15] **Logic-Level Evaluation of Logical Table-to-Text Generation** CoNLL'26
L. Trigg, D.F. Hougen, A. Bilal

[PC16] **ITDPDM: Information-Theoretic Discrete Poisson Diffusion Model** NeurIPS'25
S. Bhattacharya, A.R. Gorle, A. Bilal, C. Ding, A.K.S. Yadav, T. Weissman

[PC17] **Meta-Thinking in LLMs via Multi-Agent Reinforcement Learning: A Survey** EAAI (Submitted)
A. Bilal, M.A. Mohsin, M. Umer, M.A.K. Bangash, M.A. Jamshed

Generative Models, Multimodal RAG, and AI for Wireless Systems

[PC18] **Neural Gaussian Radio Fields for Channel Estimation** KDD'26
M. Umer*, M.A. Mohsin*, A. Bilal*, J.M. Cioffi

[PC19] **Channel Prediction Under Network Distribution Shift Using Continual Learning-Based Loss Regularization** ICASSP'26
M.A. Mohsin, M. Umer, A. Bilal, M.I. Qadir, M.A. Jamshed, D.F. Hougen, J.M. Cioffi

[PC20] **Conditional Prior-Based Non-Stationary Channel Estimation Using Accelerated Diffusion Models** ICASSP'26
M.A. Mohsin*, A. Bilal*, M. Umer, A. Ali, M.A. Jamshed, D.F. Hougen, J.M. Cioffi

[PC21] **Transformer-Based Sparse CSI Estimation for Non-Stationary Channels** ICC'26
M.A. Mohsin, M. Umer, A. Bilal, H. Rizwan, S. Bhattacharya, M.A. Jamshed, J.M. Cioffi

[PC22] **Canonical Optimization for MIMO MAC Design** GLOBECOM'26
M. Umer, M.A. Mohsin, A. Bilal, J.M. Cioffi

[PC23] **6G Twin: Hybrid Gaussian Radio Fields for Channel Estimation and Non-Linear Precoder Design for Radio Access Networks** arXiv'25
M.A. Mohsin, M. Umer, A. Bilal, M.A. Jamshed, D.F. Hougen, J.M. Cioffi

[PC24] **Retrieval Augmented Generation with Multi-Modal LLM Framework for Wireless Environments** ICC'25 Workshop **(Best Paper)**
M.A. Mohsin, A. Bilal, S. Bhattacharya, J.M. Cioffi

[PC25] **Continual Learning for Wireless Channel Prediction** ICML'25 Workshop
M.A. Mohsin, M. Umer, A. Bilal, M.A. Jamshed, J.M. Cioffi

[PC26] **Task and Perception-aware Distributed Source Coding for Correlated Speech under Bandwidth-constrained Channels** AAAI'25 Workshop
S. Bhattacharya, M.A. Mohsin, A. Bilal, J.M. Cioffi

[PC27] **HDRL for Spectrum Resource Optimization in Integrated Terrestrial and Non-Terrestrial Networks** AAAI'25 Workshop
M.A. Mohsin, H. Rizwan, M. Umer, S. Bhattacharya, A. Bilal, J.M. Cioffi

Computer Vision and Explainable AI

[PC28] **On Shadow Removal With Boosted Attention in a Vision Transformer** Springer ML (Submitted)
A. Bilal, A. Salman, K. Khurshid, D.F. Hougen

[PC29] **Person Identification using Gait with Fused Graph and 3D Convolutional Architectures** ACM TAIS (Submitted)
A. Bilal, A. Salman, K. Khurshid

[PC30] **LLMs for Explainable AI: A Comprehensive Survey** arXiv'25
A. Bilal, D. Ebert, B. Lin

[PC31] **Abstract – LLM for Explainable AI** IEEE DSAA'24
A. Bilal, B. Lin

TEACHING EXPERIENCE

Teaching Assistant University of Oklahoma
CS-1313: Programming for Non-majors in C *Fall 2024 – Present*

- Supervised weekly lab sessions; designed assignments, graded with clear rubrics, and led help sessions supporting students with C programming. Collaborated with Dr. Neeman to develop supplementary written technical documentation; communicated clearly with students in a collaborative classroom environment to address individual problem areas and foster analytical thinking.

REVIEWER, TALKS & OPEN SOURCE

Conference Reviewer	ICLR, ICML, NeurIPS, ICASSP, PAKDD.
Journal Reviewer	TMLR, IEEE WCM, Springer MT&A, IP&M, Aquaculture Int., IJIM, IEEE Access, ICES, ISFI.
Talks	Gave a talk on "AI in Healthcare" at Norman Regional Hospital under Dr. Lubna Mirza.
Open-Source Contributor	Contributor to MatrAIx.ai, an open-source organization building persona-driven agent and user-simulation tools; co-author of AgentPersonaBench, a benchmark for persona-driven user simulation whose dataset, environments, and evaluation code are publicly released.

HONORS AND ACHIEVEMENTS

- **Honorary Certificate of Appreciation**: IEEE Communications Society Student Competition 2025 for "Democratizing 6G: AI-Native Wireless Digital Twin for Global Digital Equity and Sustainability."
- **Best Student Presentation Runner-up Award**: IEEE DSAA'24 Student Forum.
- **Graduate Fellowship**: Awarded Gallogly College of Engineering Graduate Fellowship 2025.
- **Best Paper Award**: ICC Workshop 2025 in Montreal.
- **Student Travel Grant**: IEEE DSAA 2024 in San Diego.
- **Best Adjudged Industrial Project Award**: Final Year Project received 1st place at NUST Open House 2024.

- **UGRIP Selection**: Selected as Undergraduate Research Intern for First Cohort by MBZUAI.
- **IEEE Recognition**: Selected as an Emerging Young Researcher in the IEEE Islamabad Section.
- **Prime Minister Laptop Scheme**: Winner of scholarship program.

CERTIFICATIONS

- Google UX Professional Certificate
- Deep Learning Specialization Certificate
- The Advanced Communication Skills Course

SKILLS

Programming	Python, C/C++, Java, Embedded C, MATLAB. Proficient in scripting, algorithm design, and data structures.
Deep Learning & ML	PyTorch, TensorFlow (Keras), OpenCV, Docker, Mlflow, EC2 Instance. Strong machine learning background spanning neural network architectures, natural language processing (NLP), continuous learning, and scalable pipelines for large-scale data processing and dataset management. Proficient in Dash for interactive data analysis and visualization.
LLM & Alignment	Transformers and attention variants, supervised fine-tuning (SFT), RLHF/RLAIF, DPO and preference optimization, PPO/GRPO/RLVR, reward and process reward modelling, knowledge distillation, test-time compute scaling, discrete diffusion language models, scaling laws, and LLM evaluation and benchmarking.
Efficient LLM Systems	KV cache management (eviction, compression, quantization), sparse and long-context attention, memory optimization, speculative decoding, retrieval-augmented generation (RAG), and efficient inference and serving.
Agentic AI	LLM agents, multi-agent systems, tool use, planning, long-term agent memory, self-improving and self-evolving agents, agent communication protocols, and the Internet of Agents.
AI Safety & Security	Alignment, AGI safety, sycophancy, hallucination, calibration and uncertainty quantification, red teaming, jailbreak and prompt-injection robustness, adversarial robustness, interpretability, and trustworthy AI.
Statistics & Theory	Analytical and statistical learning theory, probabilistic modelling, information theory, and optimization methods applied to machine learning research; experienced in rigorous analysis and problem formulation from specification through evaluation.
Infrastructure & Systems	Distributed systems design, HTTP APIs, Docker-based infrastructure with security best practices, scalable deployment on cloud and edge. Familiar with development best practices, including version control, clean code, and written technical documentation; applied rapid prototyping to iterate quickly across diverse use cases.
Web Development	JavaScript (React.js), HTML/CSS, Next.js.
Tools	VS Code, Git, AutoCad, Figma, PyCharm, Raspberry Pi OS, NginX.
Soft Skills	Collaborative teamwork across cross-functional teams; communicate technical status and findings clearly to diverse audiences and stakeholders; supervised junior researchers and student groups; self-starter with attention to detail and innovative problem-solving in fast-paced environments.
Design	Figma, Adobe XD, Adobe Illustrator, Adobe Photoshop, Sketch, WordPress Theme Design.
Core Competencies	LLMs and Foundation Models, AI Alignment, Preference Optimization (RLHF, DPO, GRPO), AI Safety and Security, Agentic AI, Self-Improving Agents, KV Cache and Memory Optimization, Test-Time Compute Scaling, Reasoning, Reinforcement Learning, Signal Processing, Machine Learning (ML), Classification, Algorithms, 5G, Connectivity Technologies, Compliance, Data Analysis (pandas), Model Development (scikit-learn), Distributed AI Systems.